Genuine Multipartite Entanglement Analysis — Molecular System

MOLECULE

Benzene (C6H6)
Basis set   : cc-pVDZ
Charge   : 0
Multiplicity : 1
Total e-   : 42
n-body order : 4

GROUND STATE ENERGIES

E0 (Ha) : -230.75355805
E1 (Ha) : -230.57826157
Gap (Ha) : 0.17529648
HF (Ha) : -230.71878874

CONVEX-ROOF METHODS

Methods run   : gcg
Restarts   : 3

C_LB : coherence witness bound
C_expo: unitary opt (exp map)
C_stf : Stiefel manifold (Alg. B)
C_UB : eigendecomposition bound

ACTIVE SPACE CAS(6,6)

Frozen core orbs : 18
Frozen e-   : 36
Active e-   : 6
Active orbs   : 6
Spin-orbitals   : 12
Hilbert dim   : 400

SPIN PROPERTIES

<S2>   : 0.000000
S   : 0.0000
<Sz>   : 0.000000
Multiplicity : 1
Spin state   : Singlet (S=0)

GME CRITERIA

C_GME = sqrt(2 * min_A (1 - Tr(rho_A^2)))

Convex roof (mixed):
  C_GME(rho) = min Sum p_i C_GME(|psi_i>)

Bounds ordering:
  C_LB <= C_GME <= C_UB

DOUBLE OCCUPANCY

<D> total : 2.961506
<D> / site : 0.493584

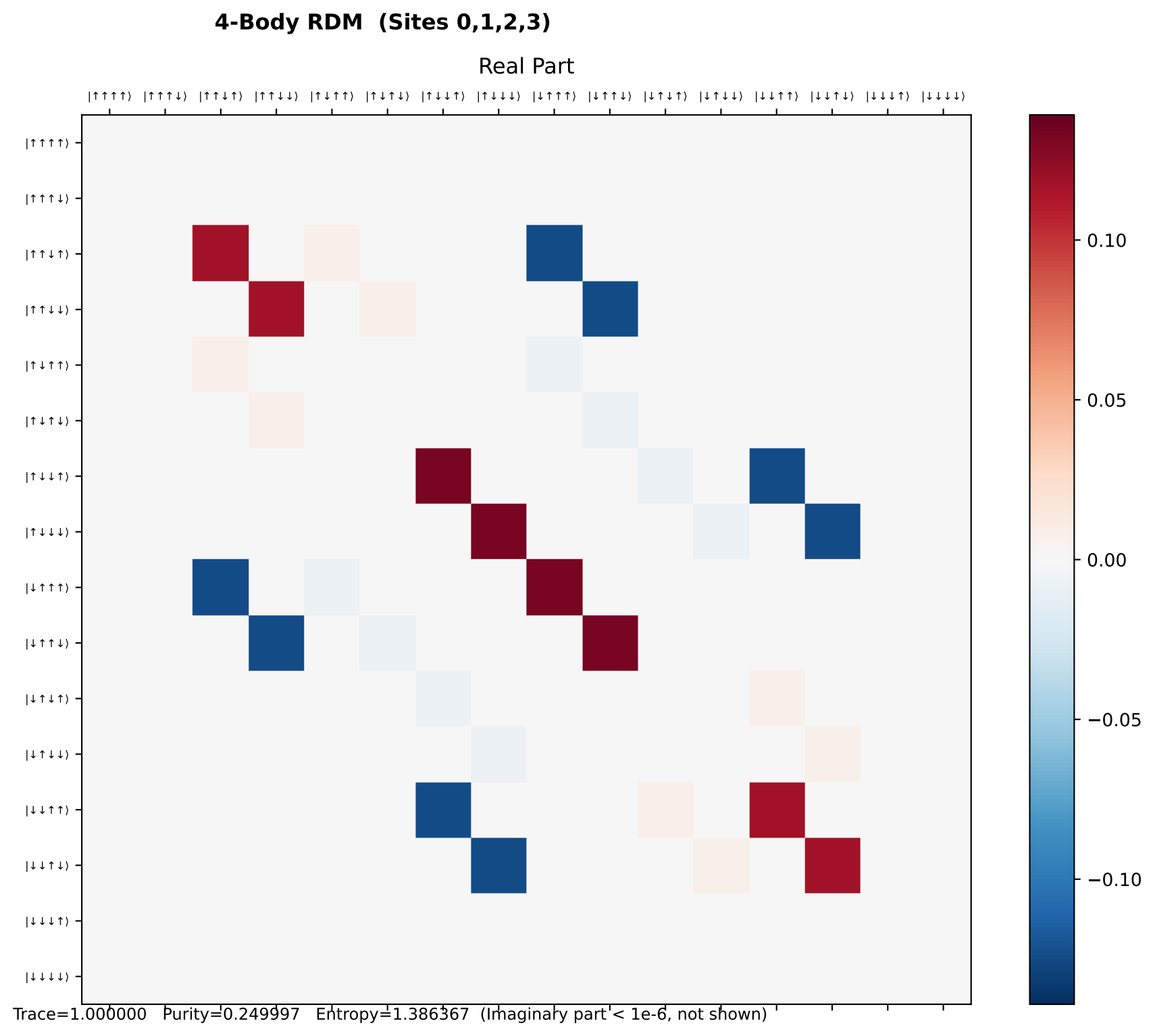
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Body order : 4
Clusters   : 15

Date: 2026-07-10 17:23

Summary — 4-Body RDMs (Benzene, CAS(6,6))

Orbitals	Purity	C_GME (pure)	C_LB	C_gcg	C_stiefel	C_UB	State
0,1,2,3	0.249997	—	0.000000	0.001243	—	0.053768	MIXED
0,1,2,4	0.249997	—	0.000000	0.001049	—	0.053767	MIXED
0,1,2,5	0.834591	—	0.471804	0.000008	—	0.090758	WEAKLY MIXED
0,1,3,4	0.249998	—	0.000000	0.001106	—	0.053768	MIXED
0,1,3,5	0.228267	—	0.000000	0.003652	—	0.374537	MIXED
0,1,4,5	0.228210	—	0.000000	0.003162	—	0.375166	MIXED
0,2,3,4	0.249998	—	0.000000	0.000890	—	0.053835	MIXED
0,2,3,5	0.228511	—	0.000000	0.003364	—	0.372263	MIXED
0,2,4,5	0.228456	—	0.000000	0.003189	—	0.372934	MIXED
0,3,4,5	0.801534	—	0.464790	0.000001	—	0.109389	WEAKLY MIXED
1,2,3,4	1.000000	—	0.530061	0.107532	—	0.107532	PURE
1,2,3,5	0.192480	—	0.000000	0.033591	—	0.150506	STRONGLY MIXED
1,2,4,5	0.191842	—	0.000000	0.027516	—	0.153835	STRONGLY MIXED
1,3,4,5	0.212280	—	0.000000	0.047973	—	0.080377	MIXED
2,3,4,5	0.212777	—	0.000000	0.040533	—	0.079753	MIXED


4-Body RDM (Sites 0,1,2,3)
Real Part
|↑↑↑↑⟩ |↑↑↑↓⟩ |↑↑↓↑⟩ |↑↑↓↓⟩ |↑↓↑↑⟩ |↑↓↑↓⟩ |↑↓↓↑⟩ |↑↓↓↓⟩ |↓↑↑↑⟩ |↓↑↑↓⟩ |↓↑↓↑⟩ |↓↑↓↓⟩ |↓↓↑↑⟩ |↓↓↑↓⟩ |↓↓↓↑⟩ |↓↓↓↓⟩
0.10
0.05
0.00
−0.05
−0.10
Trace=1.000000 Purity=0.249997 Entropy=1.386367 (Imaginary part < 1e-6, not shown)

Analysis Summary - 4-Body RDM (Sites 0,1,2,3)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.2499972350
  Von Neumann entropy  : 1.3863667067
  Eigenvalue min       : -1.280478e-17
  Eigenvalue max       : 2.499991e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 0.24999724

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

STATUS: MIXED STATE
-------------------

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

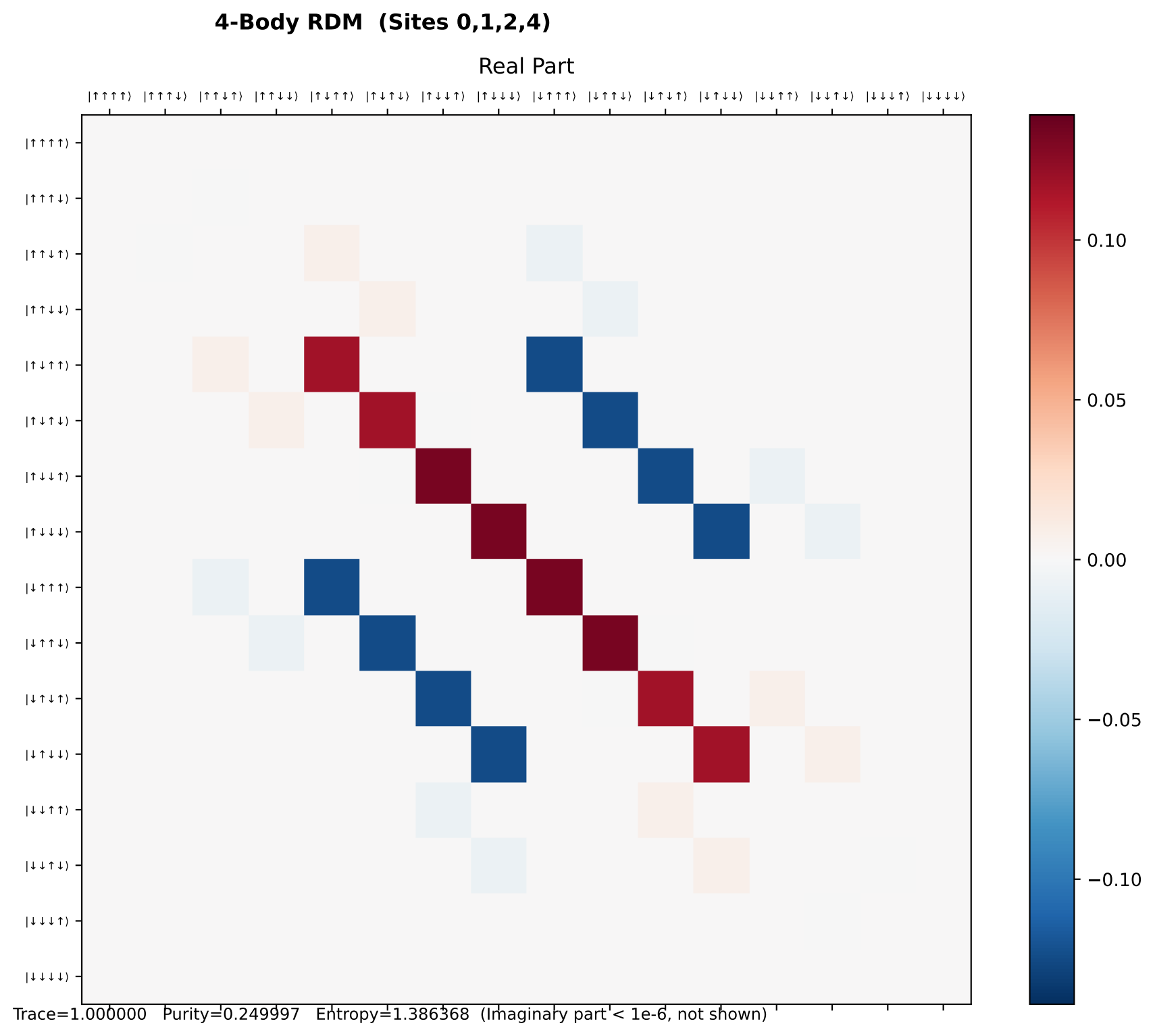
```
Convex Roof Analysis
====================================

  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = nan   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.05376789

  Status: inconclusive

  Solver source:
    –
```

```
1. p=0.249757  C_GME=0.000044
2. p=0.001479  C_GME=0.084106
3. p=0.246307  C_GME=0.000209
4. p=0.001332  C_GME=0.087845
5. p=0.182475  C_GME=0.002032
... and 5 more states
```


4-Body RDM (Sites 0,1,2,4)
Real Part
|↑↑↑↑⟩ |↑↑↑↓⟩ |↑↑↓↑⟩ |↑↑↓↓⟩ |↑↓↑↑⟩ |↑↓↑↓⟩ |↑↓↓↑⟩ |↑↓↓↓⟩ |↓↑↑↑⟩ |↓↑↑↓⟩ |↓↑↓↑⟩ |↓↑↓↓⟩ |↓↓↑↑⟩ |↓↓↑↓⟩ |↓↓↓↑⟩ |↓↓↓↓⟩
0.10
0.05
0.00
−0.05
−0.10
Trace=1.000000 Purity=0.249997 Entropy=1.386368 (Imaginary part < 1e-6, not shown)

Analysis Summary - 4-Body RDM (Sites 0,1,2,4)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.2499971871
  Von Neumann entropy  : 1.3863678625
  Eigenvalue min       : -5.083853e-18
  Eigenvalue max       : 2.499991e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 0.24999719

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

STATUS: MIXED STATE
-------------------

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

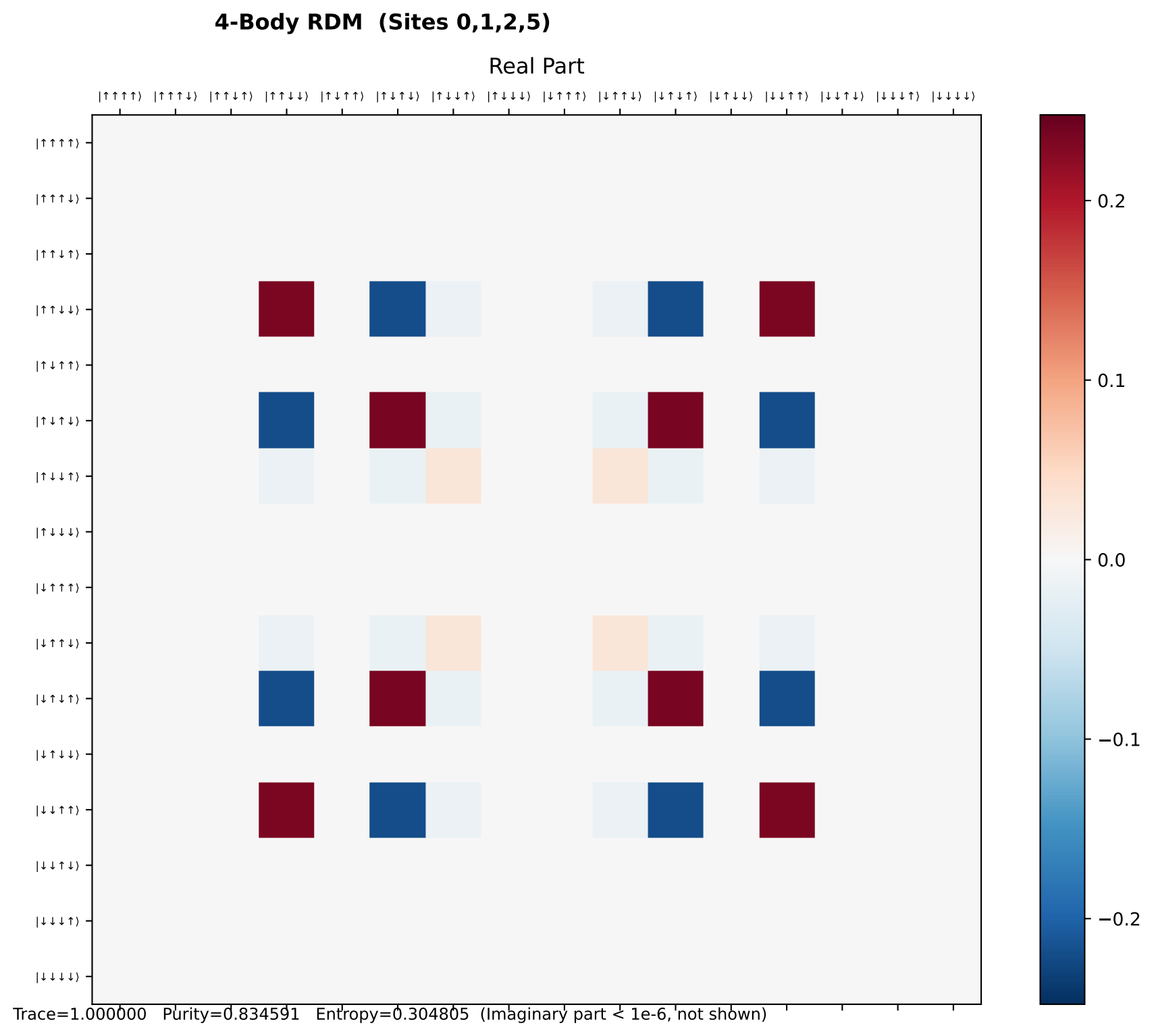
```
Convex Roof Analysis
====================================

  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = nan   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.05376745

  Status: inconclusive

  Solver source:
    –
```

```
1. p=0.241723  C_GME=0.000412
2. p=0.002689  C_GME=0.031750
3. p=0.198388  C_GME=0.001222
4. p=0.012393  C_GME=0.009732
5. p=0.000880  C_GME=0.035655
... and 5 more states
```


4-Body RDM (Sites 0,1,2,5)
Real Part
|↑↑↑↑⟩ |↑↑↑↓⟩ |↑↑↓↑⟩ |↑↑↓↓⟩ |↑↓↑↑⟩ |↑↓↑↓⟩ |↑↓↓↑⟩ |↑↓↓↓⟩ |↓↑↑↑⟩ |↓↑↑↓⟩ |↓↑↓↑⟩ |↓↑↓↓⟩ |↓↓↑↑⟩ |↓↓↑↓⟩ |↓↓↓↑⟩ |↓↓↓↓⟩
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=0.834591 Entropy=0.304805 (Imaginary part < 1e-6, not shown)

Analysis Summary - 4-Body RDM (Sites 0,1,2,5)

RDM Metrics
==========

Trace(rho)	1.0000000000
Purity Tr(rho^2)	0.8345909728
Von Neumann entropy	0.3048046140
Eigenvalue min	-3.205555e-17
Eigenvalue max	9.090177e-01
Imaginary part	< 1e-6 (not shown)

GME Analysis
==========

gamma = Tr(rho^2) = 0.83459097

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.

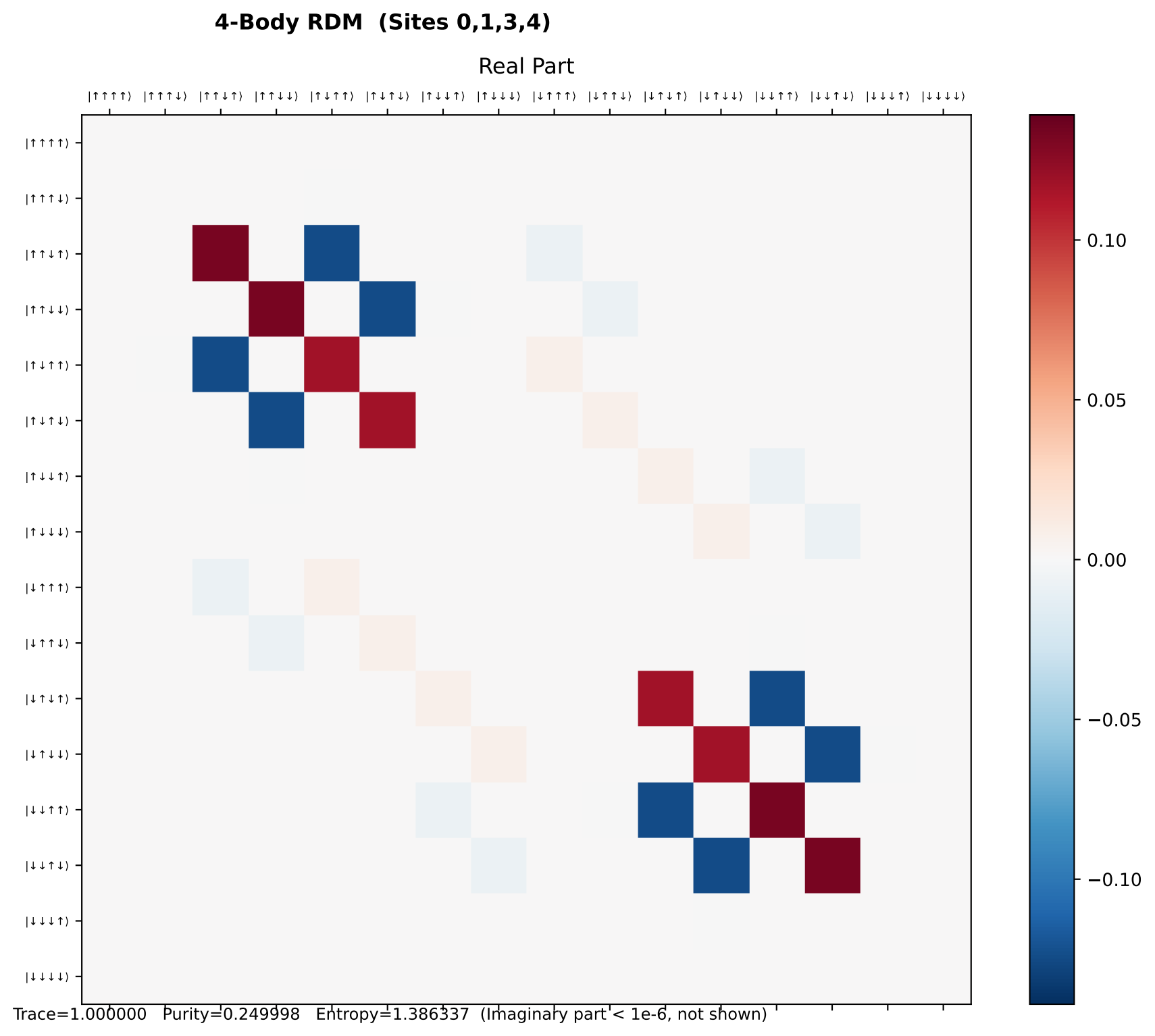
Convex Roof Analysis
==========

Lower Bounds:
C_LB = 0.47180392
C_GME (unitary optimization):
C_stiefel (unitary opt) = nan BEST
C_GME (unitary optimization - Expoenetial Map):
Upper Bound:
C_UB (eigendecomp) = 0.09075800

Status: no_GME

Solver source:
–

1. p=0.056391 C_GME=0.000110
2. p=0.025834 C_GME=0.000073
3. p=0.039110 C_GME=0.000009
4. p=0.878666 C_GME=0.000000


4-Body RDM (Sites 0,1,3,4)
Real Part
|↑↑↑↑⟩ |↑↑↑↓⟩ |↑↑↓↑⟩ |↑↑↓↓⟩ |↑↓↑↑⟩ |↑↓↑↓⟩ |↑↓↓↑⟩ |↑↓↓↓⟩ |↓↑↑↑⟩ |↓↑↑↓⟩ |↓↑↓↑⟩ |↓↑↓↓⟩ |↓↓↑↑⟩ |↓↓↑↓⟩ |↓↓↓↑⟩ |↓↓↓↓⟩
0.10
0.05
0.00
−0.05
−0.10
Trace=1.000000 Purity=0.249998 Entropy=1.386337 (Imaginary part < 1e-6, not shown)

Analysis Summary - 4-Body RDM (Sites 0,1,3,4)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.2499984586
  Von Neumann entropy  : 1.3863365325
  Eigenvalue min       : -2.695504e-17
  Eigenvalue max       : 2.499993e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 0.24999846

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

STATUS: MIXED STATE
-------------------

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

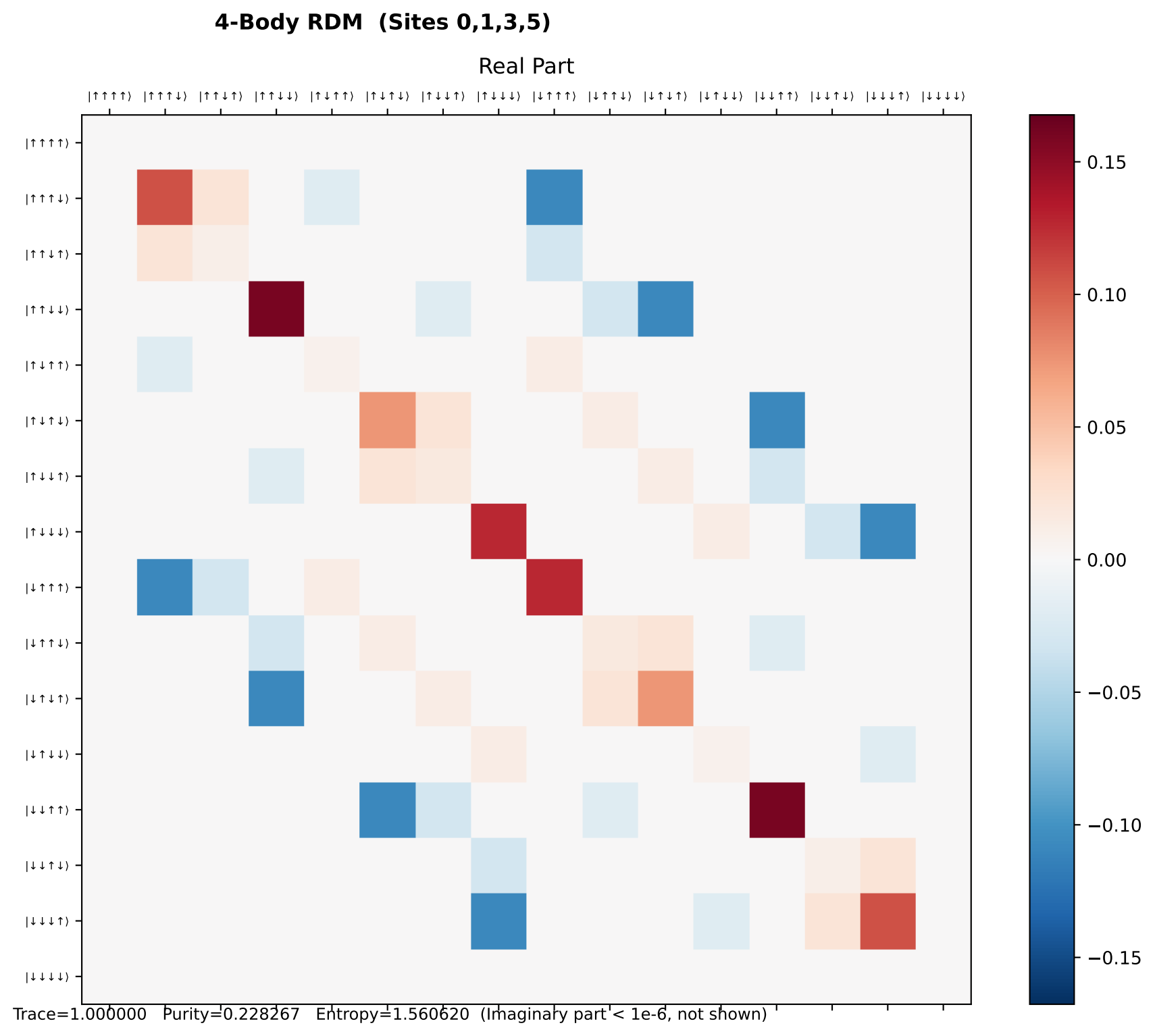
```
Convex Roof Analysis
====================================

  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = nan   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.05376780

  Status: inconclusive

  Solver source:
    –
```

```
1. p=0.199869  C_GME=0.000967
2. p=0.249296  C_GME=0.000105
3. p=0.000967  C_GME=0.093033
4. p=0.002926  C_GME=0.093452
5. p=0.012288  C_GME=0.005627
... and 5 more states
```


4-Body RDM (Sites 0,1,3,5)
Real Part
|↑↑↑↑⟩ |↑↑↑↓⟩ |↑↑↓↑⟩ |↑↑↓↓⟩ |↑↓↑↑⟩ |↑↓↑↓⟩ |↑↓↓↑⟩ |↑↓↓↓⟩ |↓↑↑↑⟩ |↓↑↑↓⟩ |↓↑↓↑⟩ |↓↑↓↓⟩ |↓↓↑↑⟩ |↓↓↑↓⟩ |↓↓↓↑⟩ |↓↓↓↓⟩
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
Trace=1.000000 Purity=0.228267 Entropy=1.560620 (Imaginary part < 1e-6, not shown)

Analysis Summary - 4-Body RDM (Sites 0,1,3,5)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.2282670771
  Von Neumann entropy  : 1.5606201050
  Eigenvalue min       : -3.523871e-17
  Eigenvalue max       : 2.499215e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 0.22826708

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

STATUS: MIXED STATE
-------------------

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

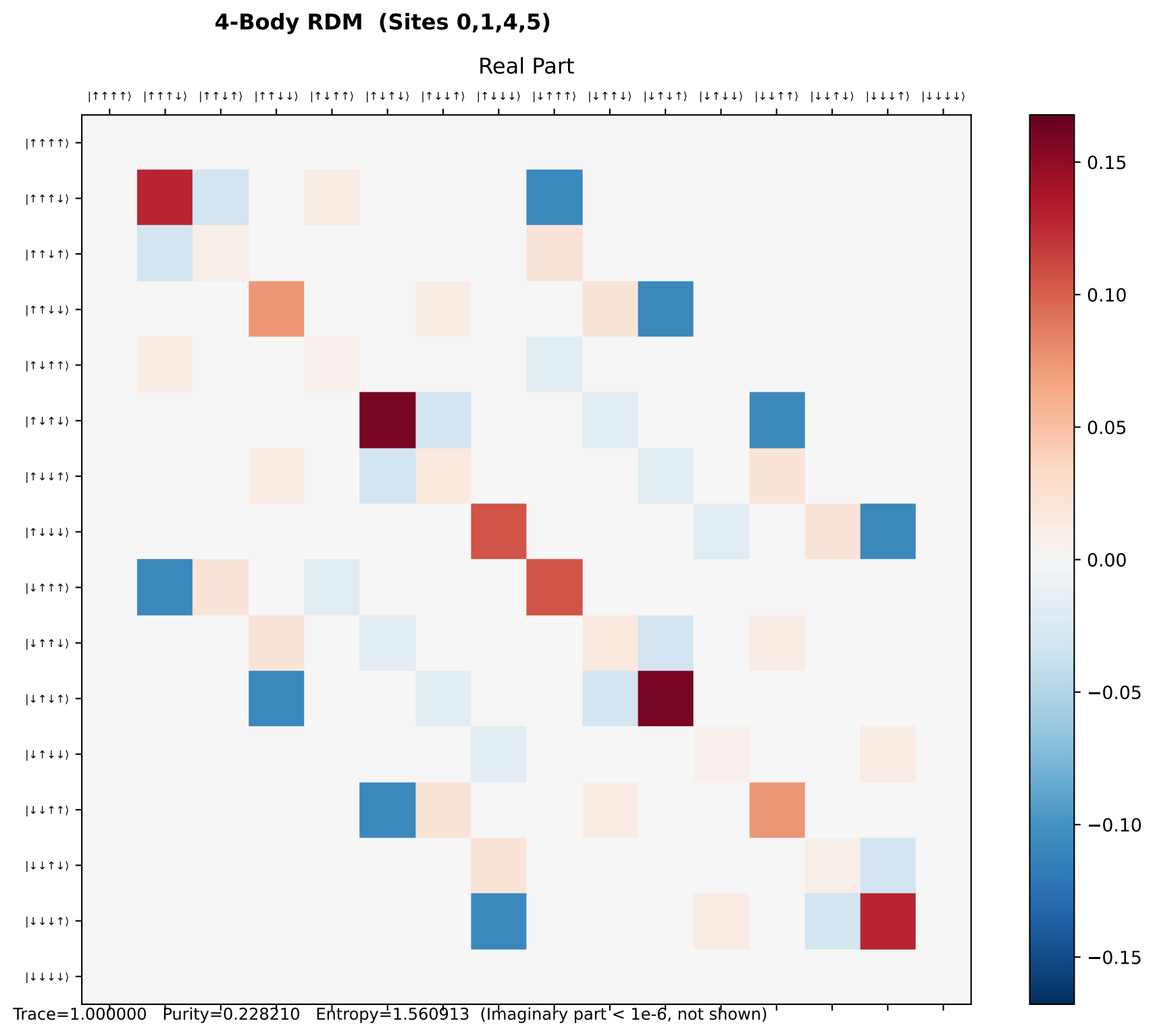
```
Convex Roof Analysis
====================================

  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = nan   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.37453748

  Status: inconclusive

  Solver source:
    –
```

```
1. p=0.159854  C_GME=0.000711
2. p=0.090377  C_GME=0.004317
3. p=0.158744  C_GME=0.004076
4. p=0.159527  C_GME=0.003050
5. p=0.000275  C_GME=0.516385
... and 5 more states
```


4-Body RDM (Sites 0,1,4,5)
Real Part
|↑↑↑↑⟩ |↑↑↑↓⟩ |↑↑↓↑⟩ |↑↑↓↓⟩ |↑↓↑↑⟩ |↑↓↑↓⟩ |↑↓↓↑⟩ |↑↓↓↓⟩ |↓↑↑↑⟩ |↓↑↑↓⟩ |↓↑↓↑⟩ |↓↑↓↓⟩ |↓↓↑↑⟩ |↓↓↑↓⟩ |↓↓↓↑⟩ |↓↓↓↓⟩
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
Trace=1.000000 Purity=0.228210 Entropy=1.560913 (Imaginary part < 1e-6, not shown)

Analysis Summary - 4-Body RDM (Sites 0,1,4,5)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.2282097687
  Von Neumann entropy  : 1.5609134306
  Eigenvalue min       : -4.029277e-17
  Eigenvalue max       : 2.499339e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 0.22820977

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

STATUS: MIXED STATE
-------------------

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

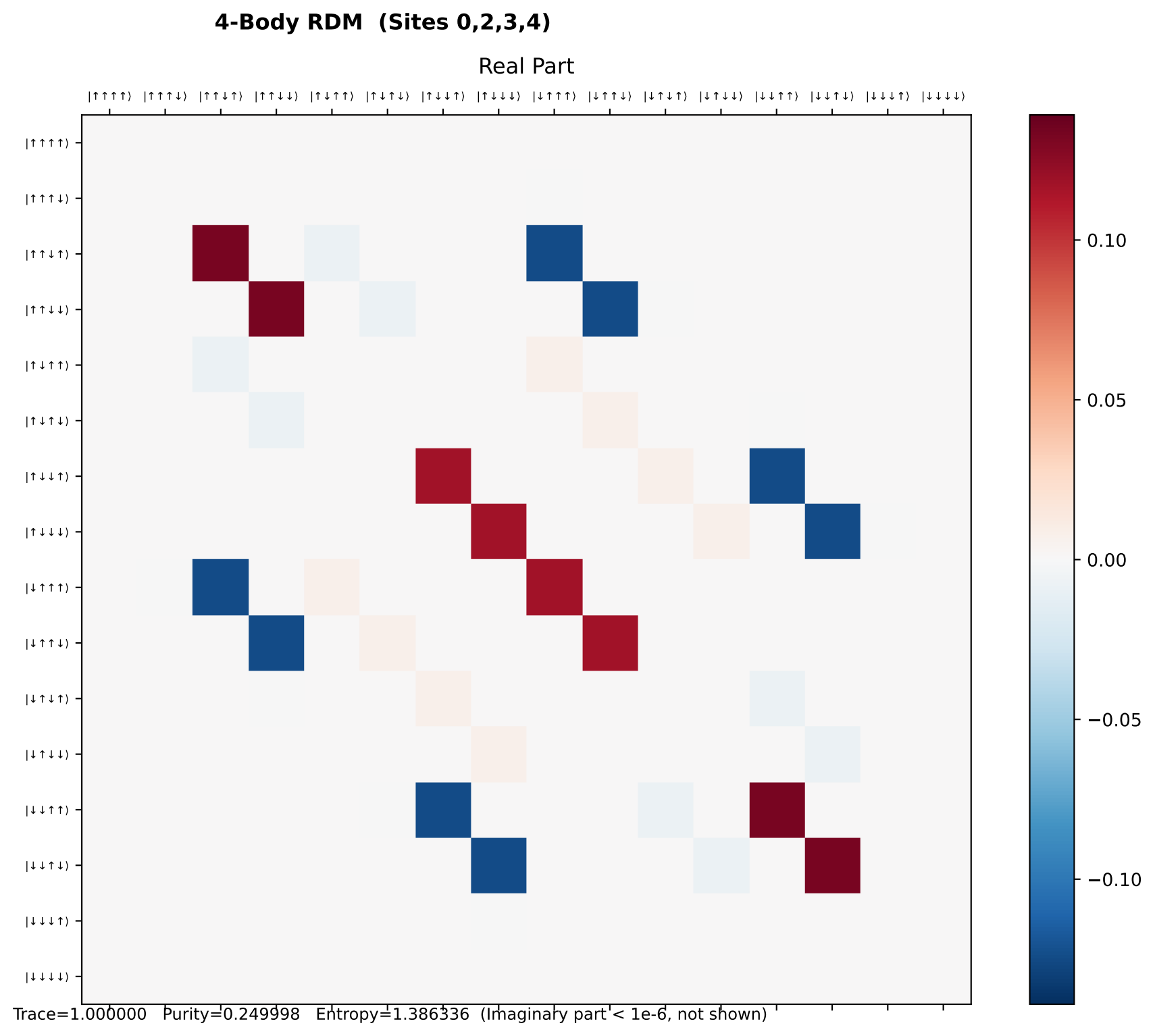
```
Convex Roof Analysis
====================================

  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = nan   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.37516586

  Status: inconclusive

  Solver source:
    –
```

```
1. p=0.084650  C_GME=0.001172
2. p=0.088007  C_GME=0.005576
3. p=0.160628  C_GME=0.001661
4. p=0.160063  C_GME=0.002409
5. p=0.161352  C_GME=0.000488
... and 5 more states
```


4-Body RDM (Sites 0,2,3,4)
Real Part
|↑↑↑↑⟩ |↑↑↑↓⟩ |↑↑↓↑⟩ |↑↑↓↓⟩ |↑↓↑↑⟩ |↑↓↑↓⟩ |↑↓↓↑⟩ |↑↓↓↓⟩ |↓↑↑↑⟩ |↓↑↑↓⟩ |↓↑↓↑⟩ |↓↑↓↓⟩ |↓↓↑↑⟩ |↓↓↑↓⟩ |↓↓↓↑⟩ |↓↓↓↓⟩
0.10
0.05
0.00
−0.05
−0.10
Trace=1.000000 Purity=0.249998 Entropy=1.386336 (Imaginary part < 1e-6, not shown)

Analysis Summary - 4-Body RDM (Sites 0,2,3,4)

RDM Metrics
==========

Trace(rho)	1.0000000000
Purity Tr(rho^2)	0.2499984844
Von Neumann entropy	1.3863358790
Eigenvalue min	-2.564762e-17
Eigenvalue max	2.499993e-01
Imaginary part	< 1e-6 (not shown)

GME Analysis
==========

gamma = Tr(rho^2) = 0.24999848

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.

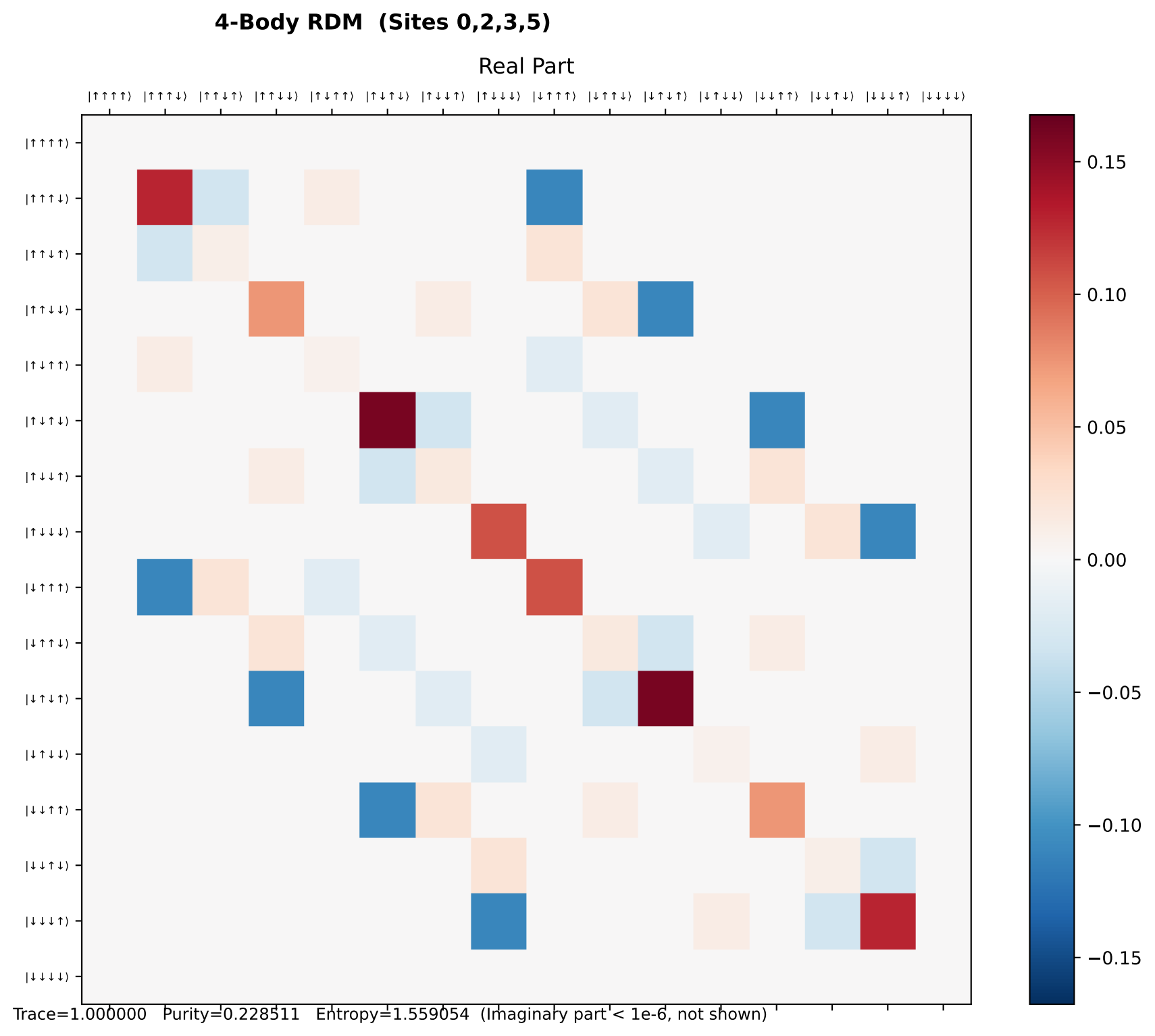
Convex Roof Analysis
==========

Lower Bounds:
C_LB = 0.00000000
C_GME (unitary optimization):
C_stiefel (unitary opt) = nan BEST
C_GME (unitary optimization - Expoenetial Map):
Upper Bound:
C_UB (eigendecomp) = 0.05383496

Status: inconclusive

Solver source:
–

1. p=0.249697 C_GME=0.000065
2. p=0.017072 C_GME=0.007113
3. p=0.001244 C_GME=0.093421
4. p=0.000058 C_GME=0.103200
5. p=0.000671 C_GME=0.106018
... and 5 more states


4-Body RDM (Sites 0,2,3,5)
Real Part
|↑↑↑↑⟩ |↑↑↑↓⟩ |↑↑↓↑⟩ |↑↑↓↓⟩ |↑↓↑↑⟩ |↑↓↑↓⟩ |↑↓↓↑⟩ |↑↓↓↓⟩ |↓↑↑↑⟩ |↓↑↑↓⟩ |↓↑↓↑⟩ |↓↑↓↓⟩ |↓↓↑↑⟩ |↓↓↑↓⟩ |↓↓↓↑⟩ |↓↓↓↓⟩
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
Trace=1.000000 Purity=0.228511 Entropy=1.559054 (Imaginary part < 1e-6, not shown)

Analysis Summary - 4-Body RDM (Sites 0,2,3,5)

RDM Metrics
==========

Trace(rho)	1.0000000000
Purity Tr(rho^2)	0.2285113709
Von Neumann entropy	1.5590535668
Eigenvalue min	-2.557186e-17
Eigenvalue max	2.499308e-01
Imaginary part	< 1e-6 (not shown)

GME Analysis
==========

gamma = Tr(rho^2) = 0.22851137

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.

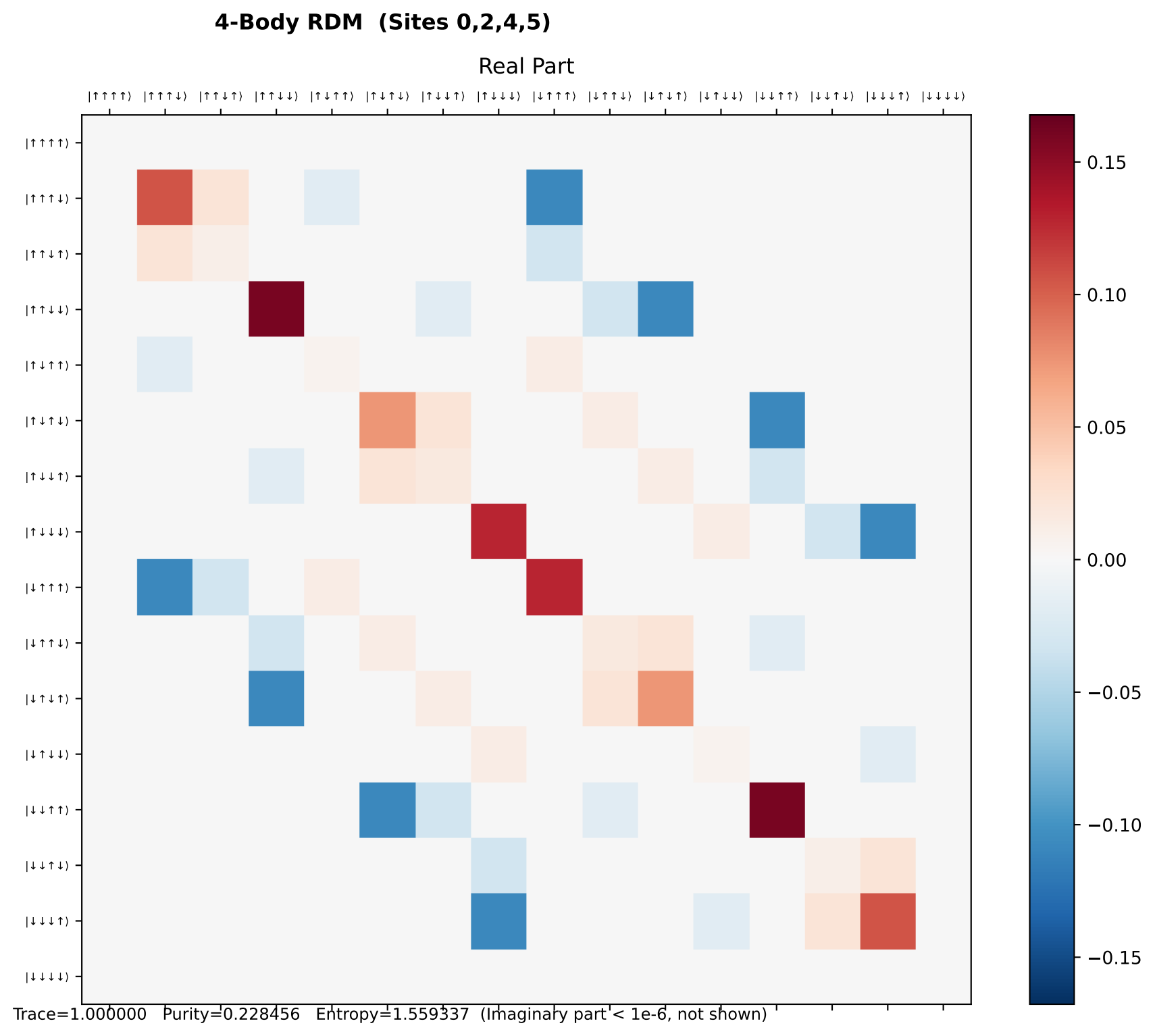
Convex Roof Analysis
==========

Lower Bounds:
  C_LB = 0.00000000
C_GME (unitary optimization):
  C_stiefel (unitary opt) = nan BEST
C_GME (unitary optimization - Expoenetial Map):
Upper Bound:
  C_UB (eigendecomp) = 0.37226266

Status: inconclusive

Solver source:
  –

1. p=0.093284 C_GME=0.004183
2. p=0.092497 C_GME=0.003907
3. p=0.093186 C_GME=0.006064
4. p=0.097555 C_GME=0.002572
5. p=0.117466 C_GME=0.002590
... and 5 more states


4-Body RDM (Sites 0,2,4,5)
Real Part
|↑↑↑↑⟩ |↑↑↑↓⟩ |↑↑↓↑⟩ |↑↑↓↓⟩ |↑↓↑↑⟩ |↑↓↑↓⟩ |↑↓↓↑⟩ |↑↓↓↓⟩ |↓↑↑↑⟩ |↓↑↑↓⟩ |↓↑↓↑⟩ |↓↑↓↓⟩ |↓↓↑↑⟩ |↓↓↑↓⟩ |↓↓↓↑⟩ |↓↓↓↓⟩
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
Trace=1.000000 Purity=0.228456 Entropy=1.559337 (Imaginary part < 1e-6, not shown)

Analysis Summary - 4-Body RDM (Sites 0,2,4,5)

RDM Metrics
==========

Trace(rho)	1.0000000000
Purity Tr(rho^2)	0.2284561949
Von Neumann entropy	1.5593374628
Eigenvalue min	-4.007288e-17
Eigenvalue max	2.499424e-01
Imaginary part	< 1e-6 (not shown)

GME Analysis
==========

gamma = Tr(rho^2) = 0.22845619

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.

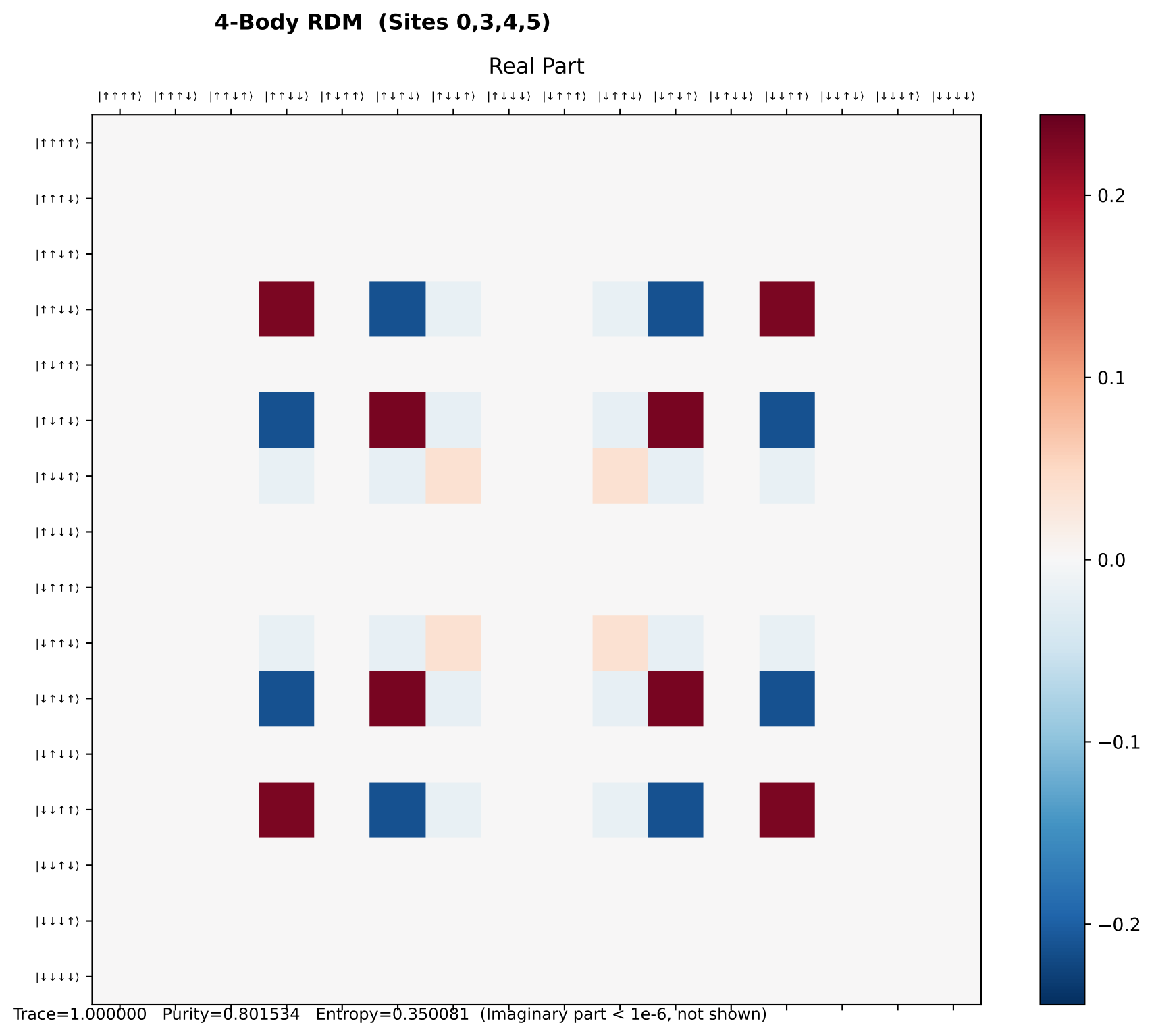
Convex Roof Analysis
==========

Lower Bounds:
C_LB = 0.00000000
C_GME (unitary optimization):
C_stiefel (unitary opt) = nan BEST
C_GME (unitary optimization - Expoenetial Map):
Upper Bound:
C_UB (eigendecomp) = 0.37293380

Status: inconclusive

Solver source:
–

1. p=0.069388 C_GME=0.006572
2. p=0.173145 C_GME=0.002894
3. p=0.172380 C_GME=0.002191
4. p=0.079702 C_GME=0.002336
5. p=0.066367 C_GME=0.002410
... and 5 more states


4-Body RDM (Sites 0,3,4,5)
Real Part
|↑↑↑↑⟩ |↑↑↑↓⟩ |↑↑↓↑⟩ |↑↑↓↓⟩ |↑↓↑↑⟩ |↑↓↑↓⟩ |↑↓↓↑⟩ |↑↓↓↓⟩ |↓↑↑↑⟩ |↓↑↑↓⟩ |↓↑↓↑⟩ |↓↑↓↓⟩ |↓↓↑↑⟩ |↓↓↑↓⟩ |↓↓↓↑⟩ |↓↓↓↓⟩
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=0.801534 Entropy=0.350081 (Imaginary part < 1e-6, not shown)

Analysis Summary - 4-Body RDM (Sites 0,3,4,5)

RDM Metrics
==========

Trace(rho)	1.0000000000
Purity Tr(rho^2)	0.8015339801
Von Neumann entropy	0.3500813350
Eigenvalue min	-3.997935e-17
Eigenvalue max	8.882873e-01
Imaginary part	< 1e-6 (not shown)

GME Analysis
==========

gamma = Tr(rho^2) = 0.80153398

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.

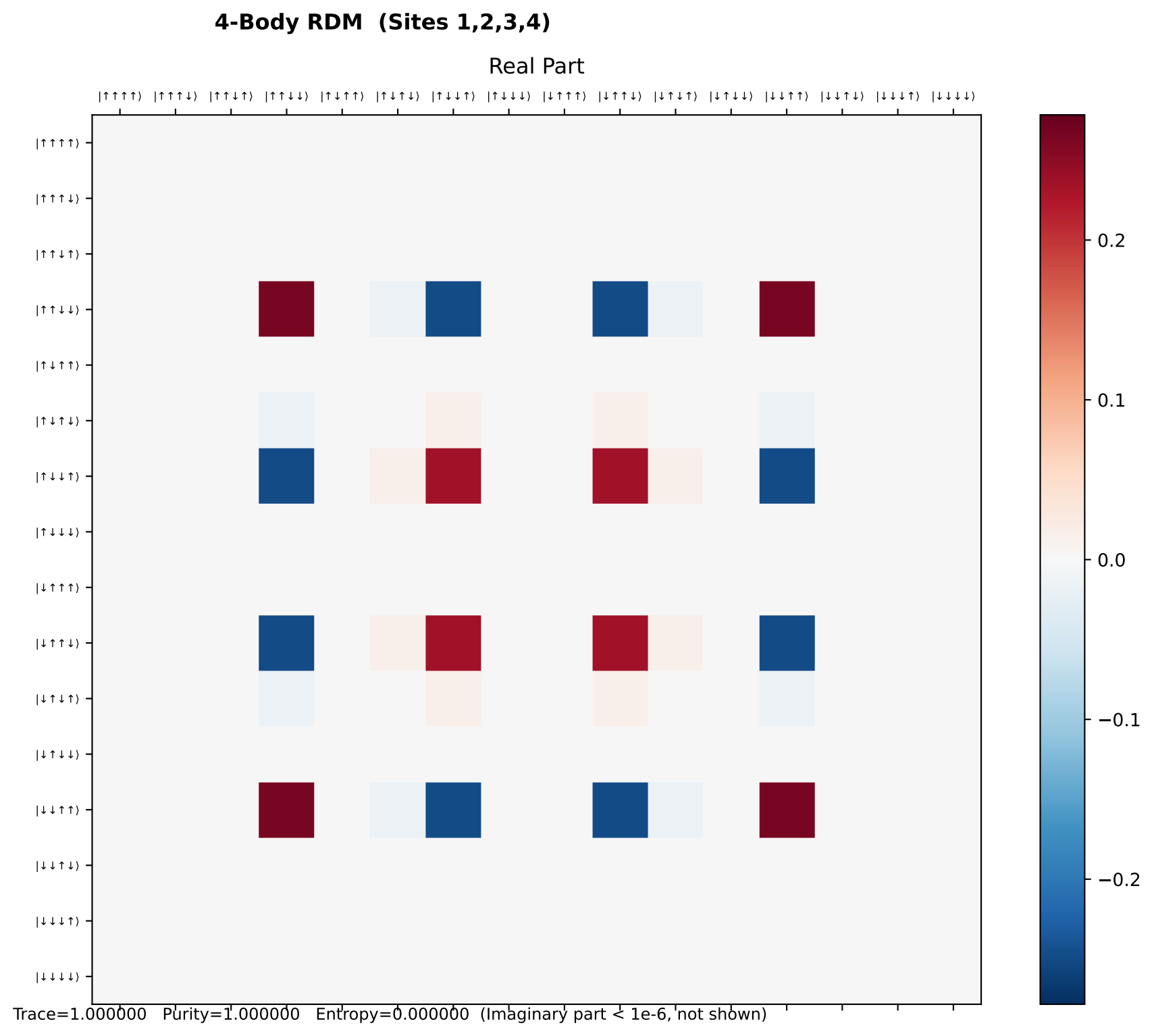
Convex Roof Analysis
==========

Lower Bounds:
C_LB = 0.46478952
C_GME (unitary optimization):
C_stiefel (unitary opt) = nan BEST
C_GME (unitary optimization - Expoenetial Map):
Upper Bound:
C_UB (eigendecomp) = 0.10938927

Status: no_GME

Solver source:
–

1. p=0.070421 C_GME=0.000004
2. p=0.030313 C_GME=0.000012
3. p=0.048238 C_GME=0.000005
4. p=0.851028 C_GME=0.000000


4-Body RDM (Sites 1,2,3,4)
Real Part
|↑↑↑↑⟩ |↑↑↑↓⟩ |↑↑↓↑⟩ |↑↑↓↓⟩ |↑↓↑↑⟩ |↑↓↑↓⟩ |↑↓↓↑⟩ |↑↓↓↓⟩ |↓↑↑↑⟩ |↓↑↑↓⟩ |↓↑↓↑⟩ |↓↑↓↓⟩ |↓↓↑↑⟩ |↓↓↑↓⟩ |↓↓↓↑⟩ |↓↓↓↓⟩
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=1.000000 Entropy=0.000000 (Imaginary part < 1e-6, not shown)

Analysis Summary - 4-Body RDM (Sites 1,2,3,4)

RDM Metrics
==========

Trace(rho)	1.0000000000
Purity Tr(rho^2)	0.9999999998
Von Neumann entropy	0.0000000019
Eigenvalue min	-1.306012e-16
Eigenvalue max	1.000000e+00
Imaginary part	< 1e-6 (not shown)

GME Analysis
==========

gamma = Tr(rho^2) = 1.00000000

A	B	Tr(rho_A^2)
(1,)	(2, 3, 4)	0.50000000
(1, 2)	(3, 4)	0.25271107
(1, 3)	(2, 4)	0.99421839
(1, 4)	(2, 3)	0.25307054
(1, 2, 3)	(4,)	0.50000000
(1, 2, 4)	(3,)	0.50000000
(1, 3, 4)	(2,)	0.50000000

STATUS: GME EXISTS

C_GME = 0.10753238
GME confirmed across ALL cuts.

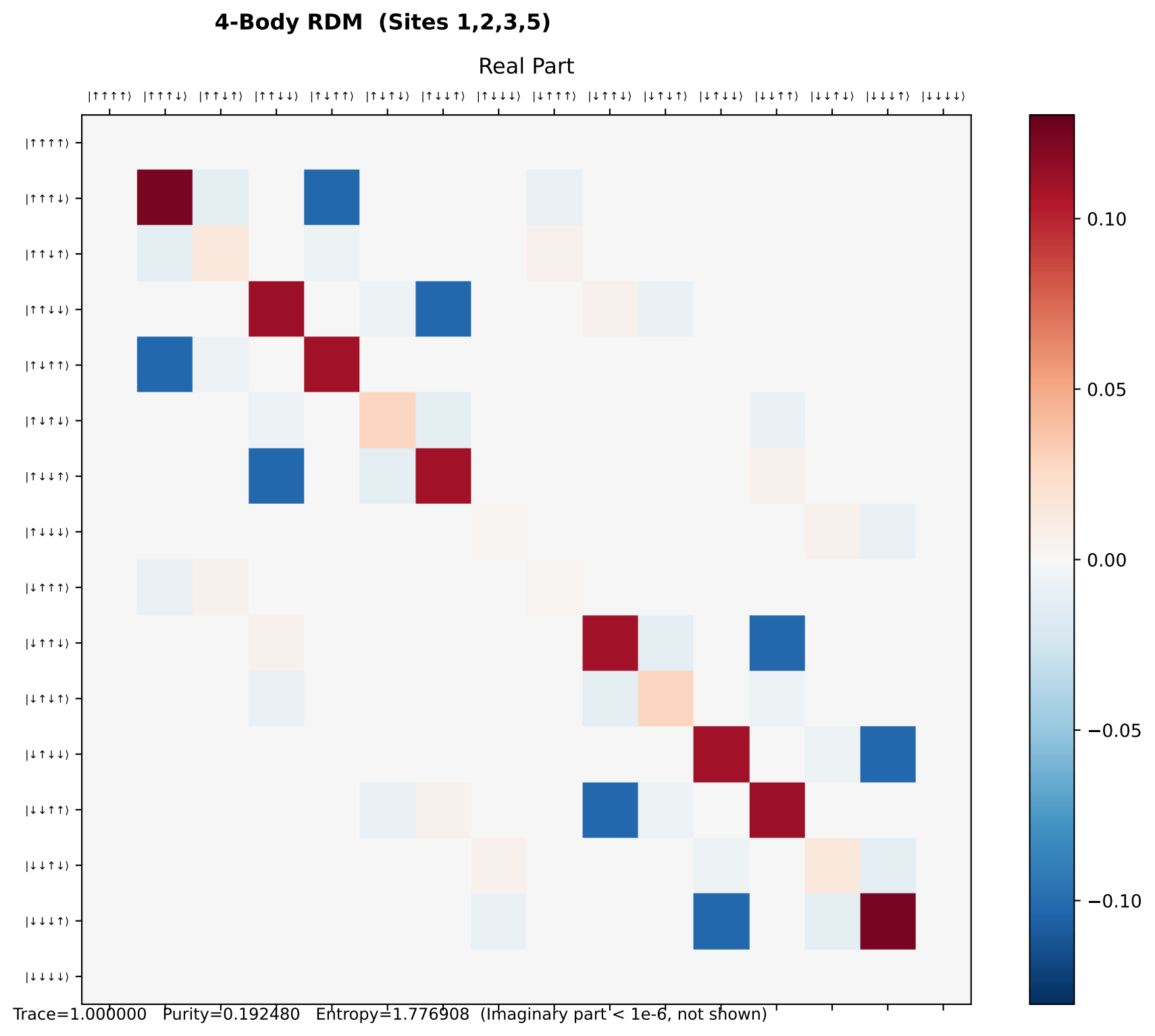
Convex Roof Analysis
==========

Lower Bounds:
  C_LB   = 0.53006144
C_GME (unitary optimization):
  C_stiefel (unitary opt) = nan   BEST
C_GME (unitary optimization - Expoenetial Map):
Upper Bound:
  C_UB (eigendecomp) = 0.10753238

Status: GME

Solver source:
  –

1. p=0.465676 C_GME=0.107532
2. p=0.534324 C_GME=0.107532


4-Body RDM (Sites 1,2,3,5)
Real Part
|↑↑↑↑⟩ |↑↑↑↓⟩ |↑↑↓↑⟩ |↑↑↓↓⟩ |↑↓↑↑⟩ |↑↓↑↓⟩ |↑↓↓↑⟩ |↑↓↓↓⟩ |↓↑↑↑⟩ |↓↑↑↓⟩ |↓↑↓↑⟩ |↓↑↓↓⟩ |↓↓↑↑⟩ |↓↓↑↓⟩ |↓↓↓↑⟩ |↓↓↓↓⟩
0.10
0.05
0.00
−0.05
−0.10
Trace=1.000000 Purity=0.192480 Entropy=1.776908 (Imaginary part < 1e-6, not shown)

Analysis Summary - 4-Body RDM (Sites 1,2,3,5)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.1924804145
  Von Neumann entropy  : 1.7769084571
  Eigenvalue min       : -2.654149e-17
  Eigenvalue max       : 2.198107e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 0.19248041

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

STATUS: MIXED STATE
-------------------

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

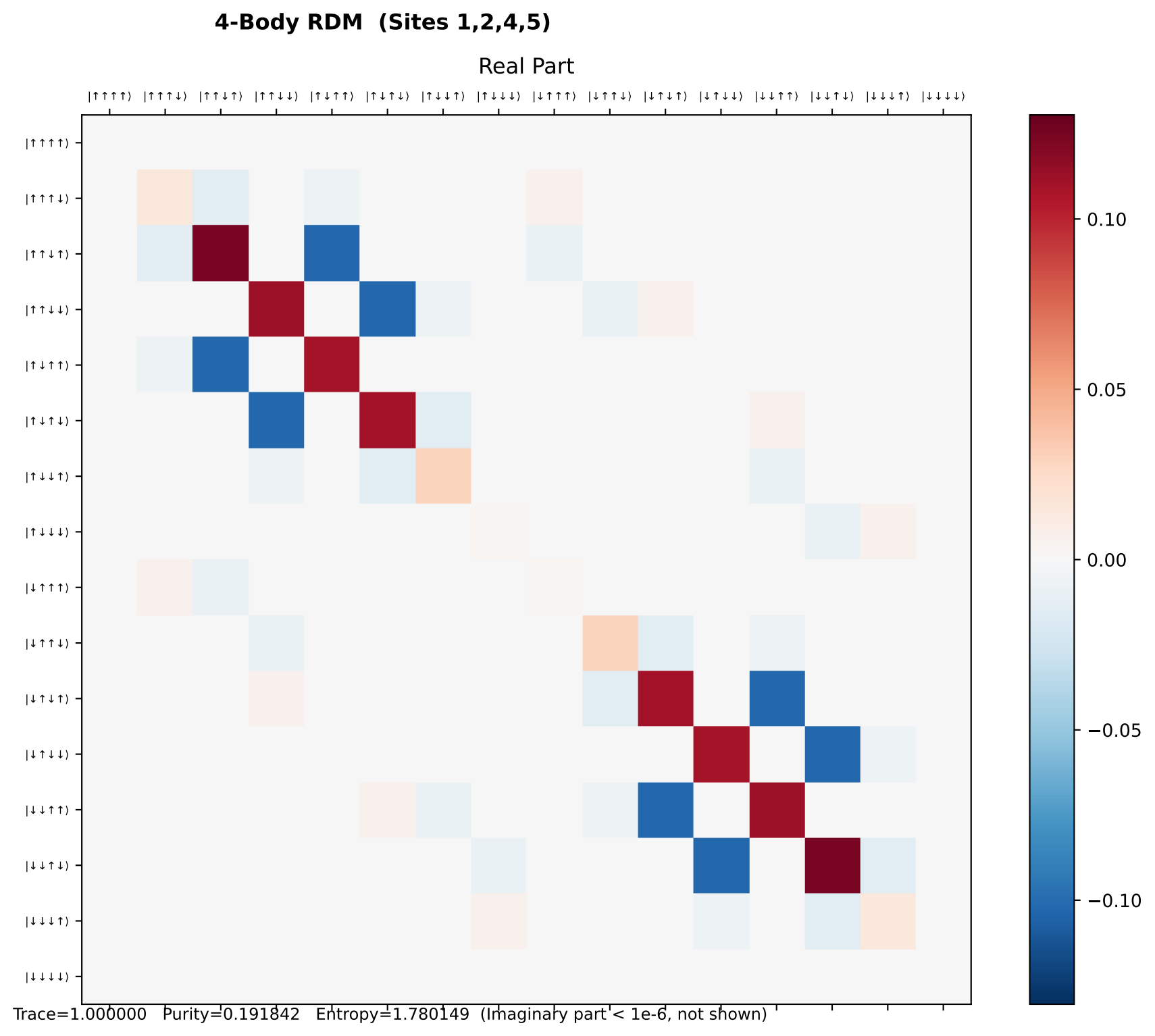
```
Convex Roof Analysis
====================================

  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = nan   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.15050630

  Status: inconclusive

  Solver source:
    –
```

```
1. p=0.043979  C_GME=0.004839
2. p=0.116763  C_GME=0.041334
3. p=0.011388  C_GME=0.153986
4. p=0.044022  C_GME=0.005596
5. p=0.043345  C_GME=0.003889
... and 5 more states
```


4-Body RDM (Sites 1,2,4,5)
Real Part
|↑↑↑↑⟩ |↑↑↑↓⟩ |↑↑↓↑⟩ |↑↑↓↓⟩ |↑↓↑↑⟩ |↑↓↑↓⟩ |↑↓↓↑⟩ |↑↓↓↓⟩ |↓↑↑↑⟩ |↓↑↑↓⟩ |↓↑↓↑⟩ |↓↑↓↓⟩ |↓↓↑↑⟩ |↓↓↑↓⟩ |↓↓↓↑⟩ |↓↓↓↓⟩
0.10
0.05
0.00
−0.05
−0.10
Trace=1.000000 Purity=0.191842 Entropy=1.780149 (Imaginary part < 1e-6, not shown)

Analysis Summary - 4-Body RDM (Sites 1,2,4,5)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.1918423382
  Von Neumann entropy  : 1.7801492269
  Eigenvalue min       : -4.908832e-17
  Eigenvalue max       : 2.194390e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 0.19184234

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

STATUS: MIXED STATE
-------------------

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

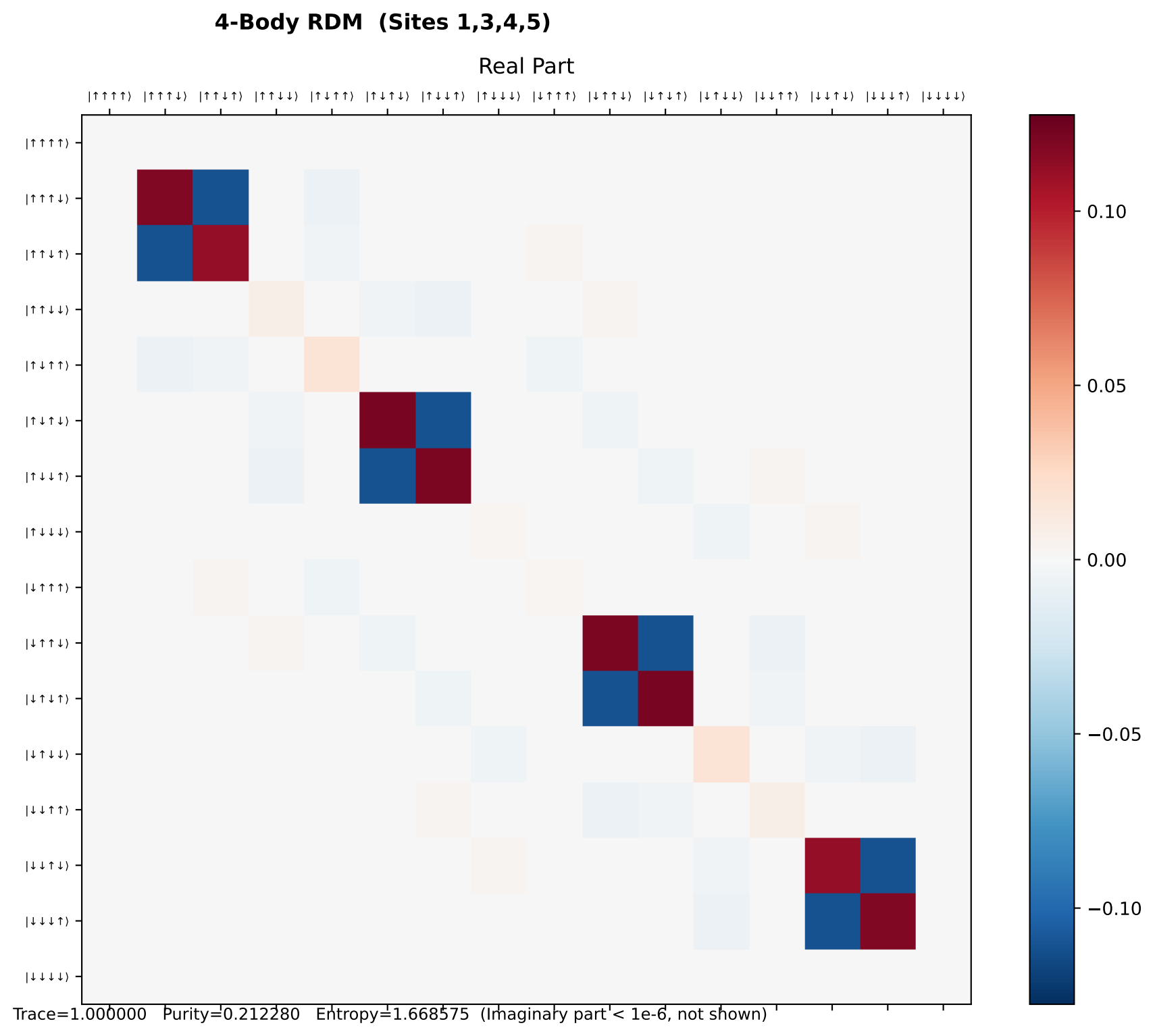
```
Convex Roof Analysis
====================================

  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = nan   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.15383465

  Status: inconclusive

  Solver source:
    –
```

```
1. p=0.035720  C_GME=0.003451
2. p=0.012883  C_GME=0.047168
3. p=0.167531  C_GME=0.006698
4. p=0.044777  C_GME=0.001238
5. p=0.041896  C_GME=0.001576
... and 5 more states
```


4-Body RDM (Sites 1,3,4,5)
Real Part
|↑↑↑↑⟩ |↑↑↑↓⟩ |↑↑↓↑⟩ |↑↑↓↓⟩ |↑↓↑↑⟩ |↑↓↑↓⟩ |↑↓↓↑⟩ |↑↓↓↓⟩ |↓↑↑↑⟩ |↓↑↑↓⟩ |↓↑↓↑⟩ |↓↑↓↓⟩ |↓↓↑↑⟩ |↓↓↑↓⟩ |↓↓↓↑⟩ |↓↓↓↓⟩
0.10
0.05
0.00
−0.05
−0.10
Trace=1.000000 Purity=0.212280 Entropy=1.668575 (Imaginary part < 1e-6, not shown)

Analysis Summary - 4-Body RDM (Sites 1,3,4,5)

```
RDM Metrics
==================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.2122800809
  Von Neumann entropy  : 1.6685754151
  Eigenvalue min       : -2.046097e-17
  Eigenvalue max       : 2.380752e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
==================================

  gamma = Tr(rho^2) = 0.21228008

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

STATUS: MIXED STATE
-------------------

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

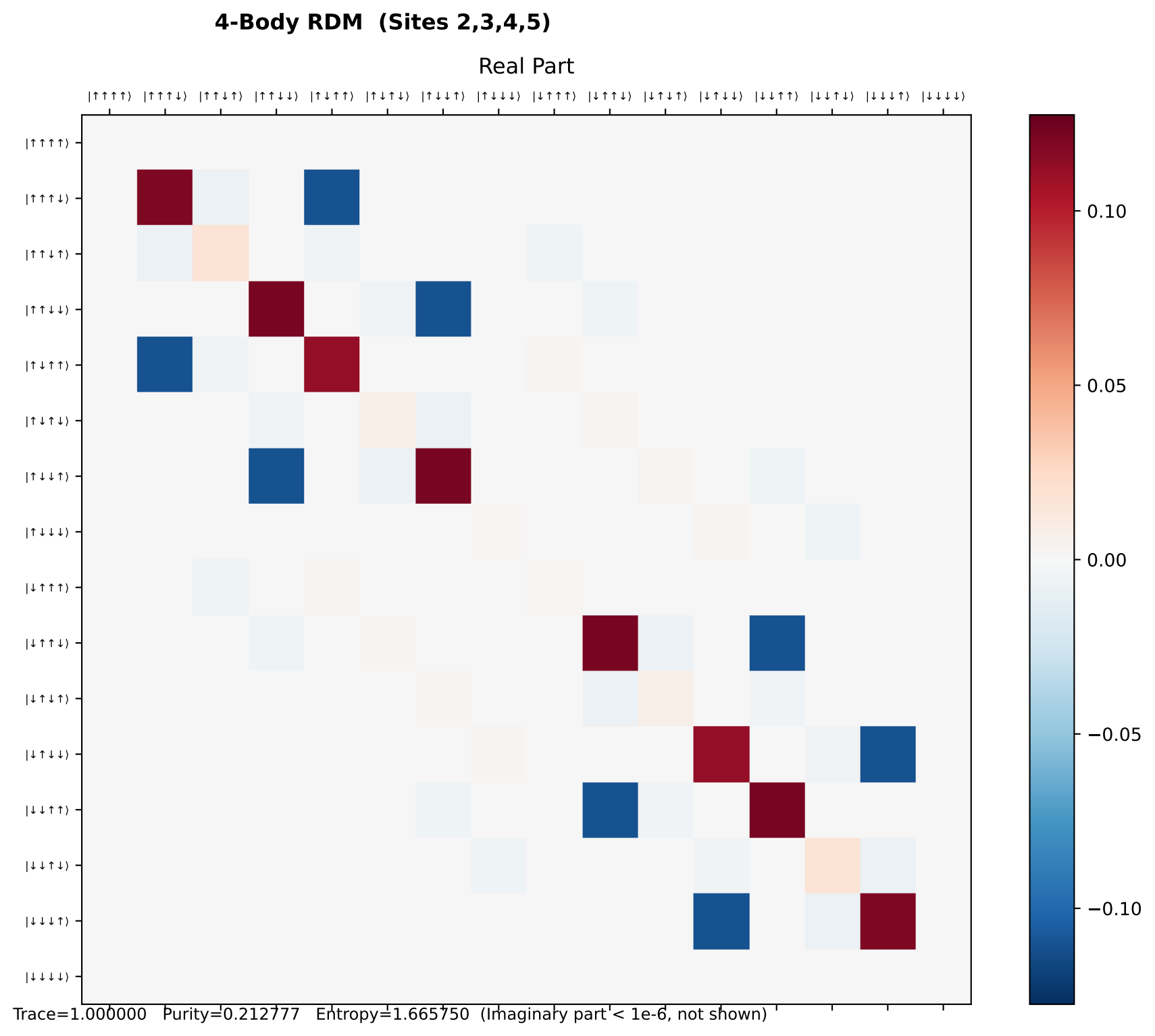
```
Convex Roof Analysis
==================================

  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = nan   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.08037657

  Status: inconclusive

  Solver source:
    –
```

```
1. p=0.231783  C_GME=0.019815
2. p=0.012351  C_GME=0.412745
3. p=0.018041  C_GME=0.127939
4. p=0.225156  C_GME=0.027416
5. p=0.223292  C_GME=0.029653
... and 5 more states
```


4-Body RDM (Sites 2,3,4,5)
Real Part
|↑↑↑↑⟩ |↑↑↑↓⟩ |↑↑↓↑⟩ |↑↑↓↓⟩ |↑↓↑↑⟩ |↑↓↑↓⟩ |↑↓↓↑⟩ |↑↓↓↓⟩ |↓↑↑↑⟩ |↓↑↑↓⟩ |↓↑↓↑⟩ |↓↑↓↓⟩ |↓↓↑↑⟩ |↓↓↑↓⟩ |↓↓↓↑⟩ |↓↓↓↓⟩
0.10
0.05
0.00
−0.05
−0.10
Trace=1.000000 Purity=0.212777 Entropy=1.665750 (Imaginary part < 1e-6, not shown)

Analysis Summary - 4-Body RDM (Sites 2,3,4,5)

RDM Metrics
==========

Trace(rho)	1.0000000000
Purity Tr(rho^2)	0.2127765290
Von Neumann entropy	1.6657503887
Eigenvalue min	-4.338199e-17
Eigenvalue max	2.381848e-01
Imaginary part	< 1e-6 (not shown)

GME Analysis
==========

gamma = Tr(rho^2) = 0.21277653

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.

Convex Roof Analysis
==========

Lower Bounds:
C_LB = 0.00000000
C_GME (unitary optimization):
C_stiefel (unitary opt) = nan BEST
C_GME (unitary optimization - Expoenetial Map):
Upper Bound:
C_UB (eigendecomp) = 0.07975275

Status: inconclusive

Solver source:
–

1. p=0.019238 C_GME=0.477869
2. p=0.223318 C_GME=0.001115
3. p=0.024852 C_GME=0.220327
4. p=0.173644 C_GME=0.036119
5. p=0.019104 C_GME=0.381255
... and 5 more states